pfi

Python for Fixed Income

COURSE 7 · NLP, RAG, AND LLMS

RAG over an Indenture

WEEK 3 · DAY 4

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Today: retrieval, citations, a judge

`question` → `retrieve` → `rag_messages` → `ask` → `judge_answer`

Today

1. Why not send the whole indenture
2. Day 6's chunks again; retrieve and rag_messages
3. Ten questions with their answers, decided first
4. Retrieval scored without the model
5. Answers with citations; the judge; grounding, not truth
6. NOT IN TEXT and the guardrail; numpy, no vector database

Saved responses of 2026-10-08, through the OpenAI route (gpt-4.1-mini-2025-04-14).

The rules for real documents

Every prompt asks what the indenture says, from the chunks sent with it

Allowed names

the saved response

the model returned

the indenture states

passed the source check

Never called

the model's opinion or view

what the model believes

loose, tight, weak, strong

a credit view or recommendation

One cell, marked adversarial, asks for an opinion on purpose: the system prompt refuses it.

Why not send the whole indenture

88,274

prompt tokens for one question (63,595 words)

33.7%

of the free model's 262,144 tokens

US$0.0088358

a request on the paid fallback's list price

Cost	every question pays for the whole document
Checking	a number could come from any of 107 sections: hard to check
Citations	five labelled chunks: the citation points to text you can read

The course's probe of 2026-10-07: accepted by the free model, the trustee named.

How retrieval works

The problem	An indenture does not fit a request; all of it costs tokens each time
Chunks	About 180 words each, turned into vectors by the same model
The match	The chunks closest to the question by cosine are retrieved
What you control	Chunk size, overlap, and k
What comes out	A prompt of a few hundred words, each chunk labelled

Retrieval and the prompt

Day 6's chunks again: 107 sections, 454 chunks, 73,008 words, embedded once

retrieve: the change of control question

```
1. Section 1.01, chunk 12, score 0.6171
2. Section 4.10, chunk 3, score 0.5671
3. Section 4.17, chunk 3, score 0.5637
4. Section 4.10, chunk 4, score 0.5371
5. Section 12.14, chunk 17, score 0.5366
```

rag_messages: one message, 5,086 characters

Rules: answer only from the sections below

a label after each sentence: [Section 4.12]

otherwise reply NOT IN TEXT

Then each chunk under its label

Then the question; ask puts the system prompt first

A chunk without its section would be a quote nobody could cite.

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495.

Where the name comes from

arXiv 2005.11401: the record, not the paper

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis and eleven co-authors; submitted 2020-05-22; Accepted at NeurIPS 2020

"providing provenance for their decisions ... remain open research problems"

RAG: "pre-trained parametric and non-parametric memory for language generation"

Their second memory: "a dense vector index of Wikipedia"

Ours: one indenture's chunks in a numpy array, citations we check

Source (read 2026-10-08): https://arxiv.org/abs/2005.11401 (its arXiv API record). The paper itself was not read.

Ten questions, decided first

Id	Question	Section	Answer key
q01	What is the interest rate on the Notes?	12.14	4.750% per annum
q02	Who is the trustee under the indenture?	before 1.01	Wilmington Trust
q03	What must the company offer holders after a change of control?	4.10	101%
q04	What is the general basket in the liens covenant?	1.01	$430,000,000, 7.5%
q05	What coverage ratio must the company meet to incur debt under the ratio test?	4.12	2.0 to 1.0
q06	At what price can the company redeem the Notes in the twelve months beginning June 15, 2027?	3.07	101.583%
q07	How much can the company pay under the general basket of the restricted payments covenant?	4.15	$400,000,000
q08	What share of the holders can accelerate the Notes after an Event of Default?	6.02	30%
q09	Which covenants are suspended while the Notes have Investment Grade Status?	4.20	4.11, 4.12, 4.15
q10	How many days before a redemption date must the company send a notice of redemption?	3.03	10 days, 60 days

Written before any search or request. Two more the indenture cannot answer: leverage today, revenue last year.

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495.

How retrieval is scored

The problem	Test retrieval before any model sees it
Recall by section	Is the section a reader would cite among the chunks?
Answer key	Are the key words in the retrieved text?
What you control	The questions, their sections, and k
What comes out	Counts of ten for each check, no request sent

Retrieval scored without the model

9 of 10

recall at 5, by section: the miss is the trustee, in no chunk

5 of 10

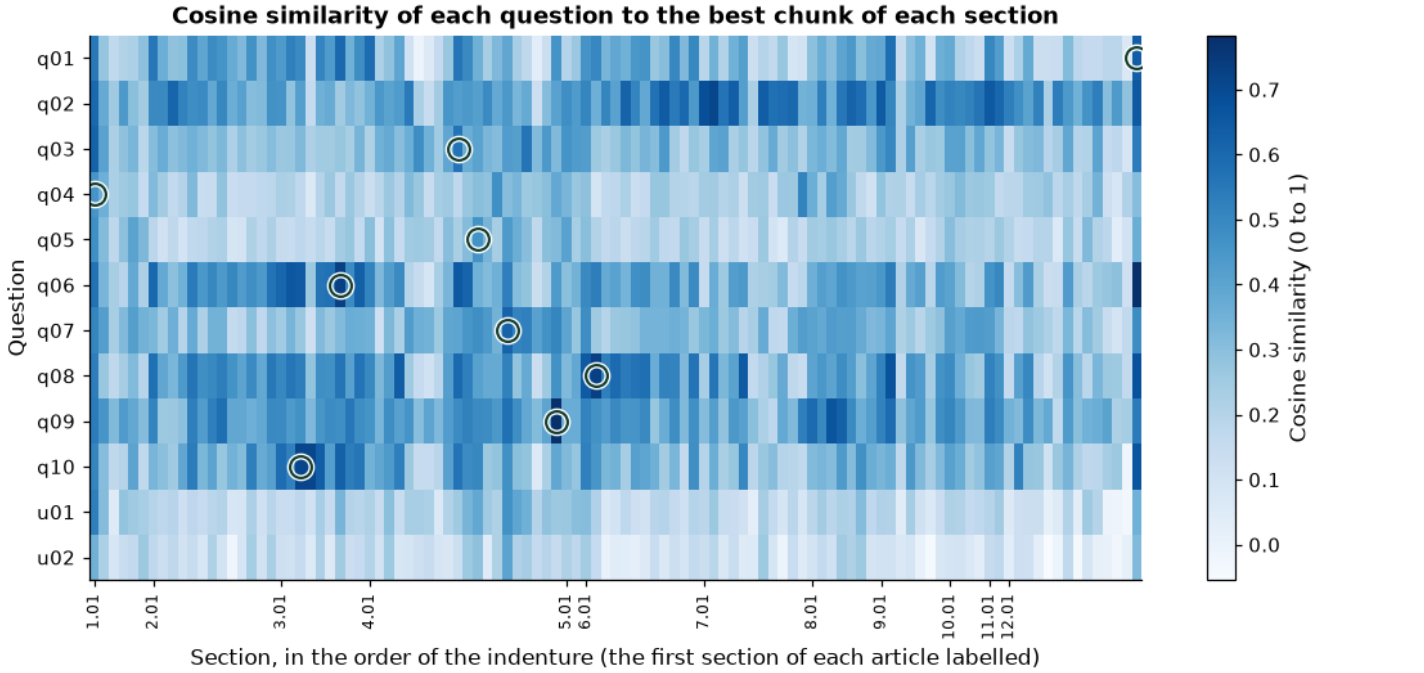
the answer key's words in the retrieved text

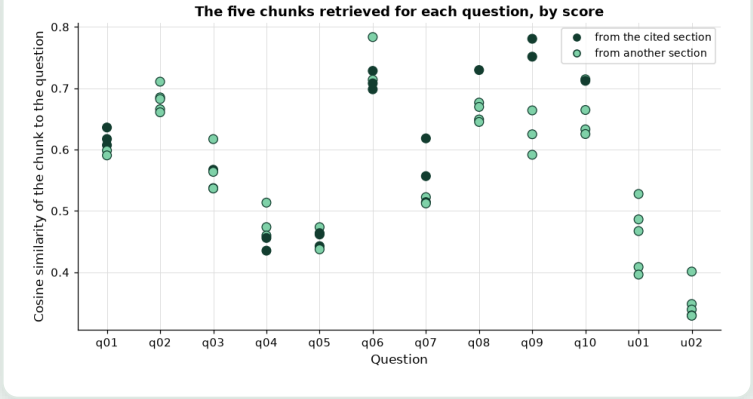
Where the covenant's first chunk ranks, of 454

```
Section 4.14: rank 184 of 454, score 0.3815  for  Can the company pledge its assets as collateral?
Section 4.10: rank 71 of 454, score 0.3779  for  What happens if the company is sold?
Section 4.14: rank 2 of 454, score 0.4736  for  What is the general basket in the liens covenant?
Section 4.10: rank 2 of 454, score 0.5671  for  What must the company offer holders after a change of control?
```

Ask in the document's vocabulary, and check recall on your own questions.

Retrieval, drawn





Darker cells: sections whose best chunk is closer to the question. Dark dots: chunks from the section a reader would cite.

A cosine says how close a chunk is to a question, not whether it answers it. Recall at 5 by section: 9 of 10.

Answers with citations

8 of 10

questions answered, each with citations

2

came back NOT IN TEXT

1,258

prompt tokens on average (1,063 to 1,510)

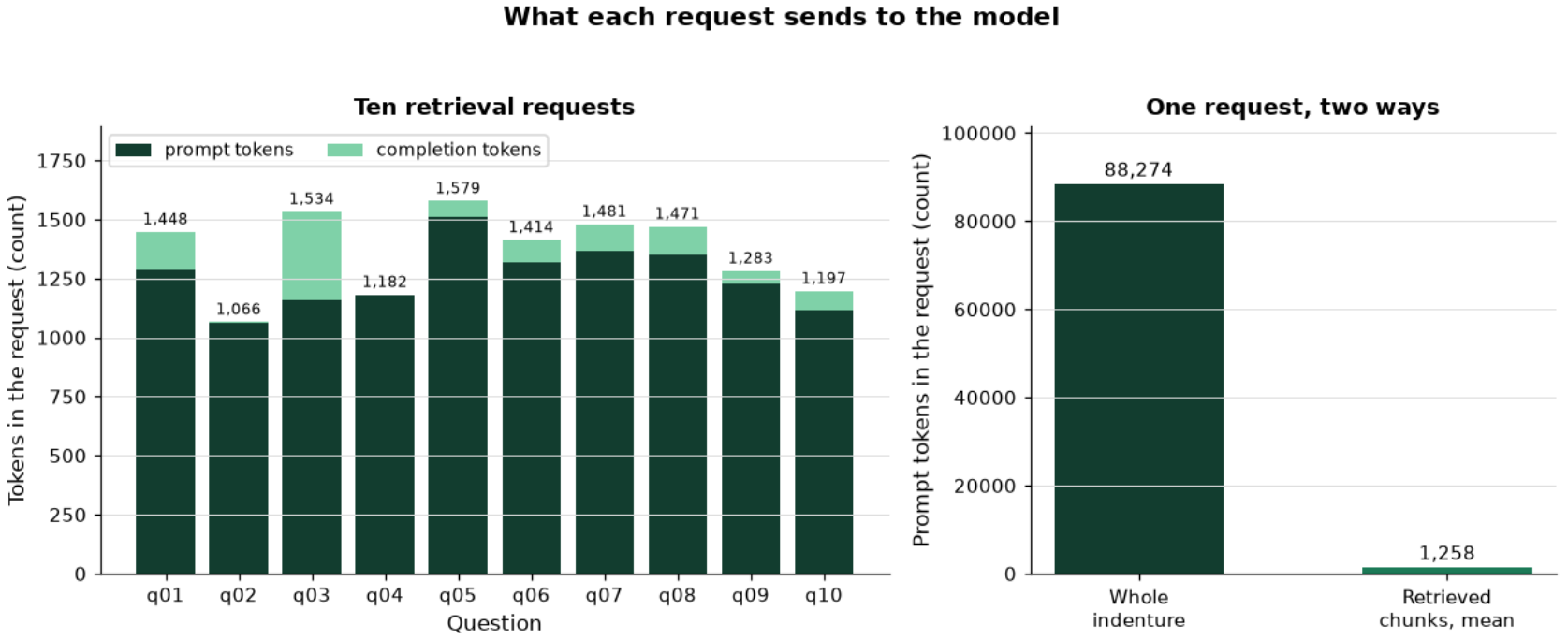
The route: OpenAI, gpt-4.1-mini-2025-04-14

On the recording day the free model answered every attempt with HTTP 429, too many requests.

Same system prompt, same message, temperature 0, seed 42; ask_rag stays for the free model.

Against 88,274 for the whole indenture: about 70 times fewer, two tokenizers, an order of size.

Tokens per request, drawn



1,258

prompt tokens, the mean retrieval request

88,274

prompt tokens, the whole indenture

Two tokenizers: read the comparison as an order of size.

How the judge works

The problem	An answer can sound grounded and not be
Rule 1	Every number in the answer is in the retrieved text
Rule 2	Every section it cites was retrieved, and one is cited
What you control	The rules and the chunks
What comes out	A verdict naming what is missing; grounding, not truth

A judge that checks grounding

Numbers	every number in the answer is written in the retrieved text, as a whole number
Citations	every cited section was retrieved, and there is at least one

8

accepted of 8 answers in the saved responses

0

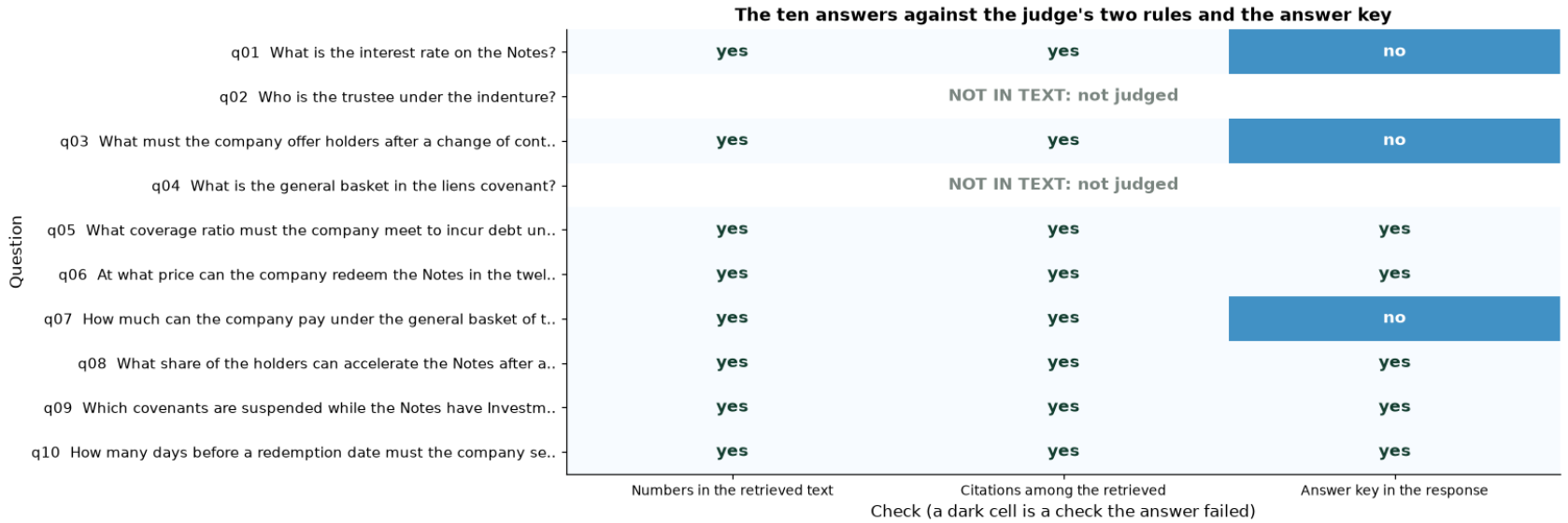
rejected

5 of 8

accepted answers that hold the answer key

Grounded is not the same as right: the three without the key answered from chunks that lack it.

The ten answers against the checks



8

accepted of 8 judged

5 of 8

accepted answers with the answer key

Grounded is not right: the dark cells of the third column sit in accepted answers.

Grounding, not truth: three planted answers

104.750% for the change of control offer

in the indenture, not in the chunks

```
rejected; missing ['104.750']
```

102.375% for the twelve months from 2027

the 2026 row of a retrieved table

```
accepted; missing []
```

The liens basket, exactly right, cited 1.01

clause (jj) was not retrieved

```
rejected; missing ['430,000,000', '7.5']
```

A filter, not a referee: it stops a number from nowhere; the answer key checks the picking.

NOT IN TEXT and the guardrail

NOT IN TEXT, counted apart

Trustee and liens basket: NOT IN TEXT

the answer was not in the five chunks

Revenue (u02): NOT IN TEXT: yes

Leverage (u01): NOT IN TEXT: no

u01 judged as an answer: accepted

The adversarial cell

"Is this a good bond to own?"

through retrieve, rag_messages, ask

REFUSED: this course asks only

what the document says

refused as asked: yes

The refusal comes from the system prompt; the rules of rag_messages say what to answer from.

Why numpy, and no vector database

454 by 384

float32, 697,344 bytes: the whole index

174,336

multiplications per search, exact

yes

the same search twice, the same rows

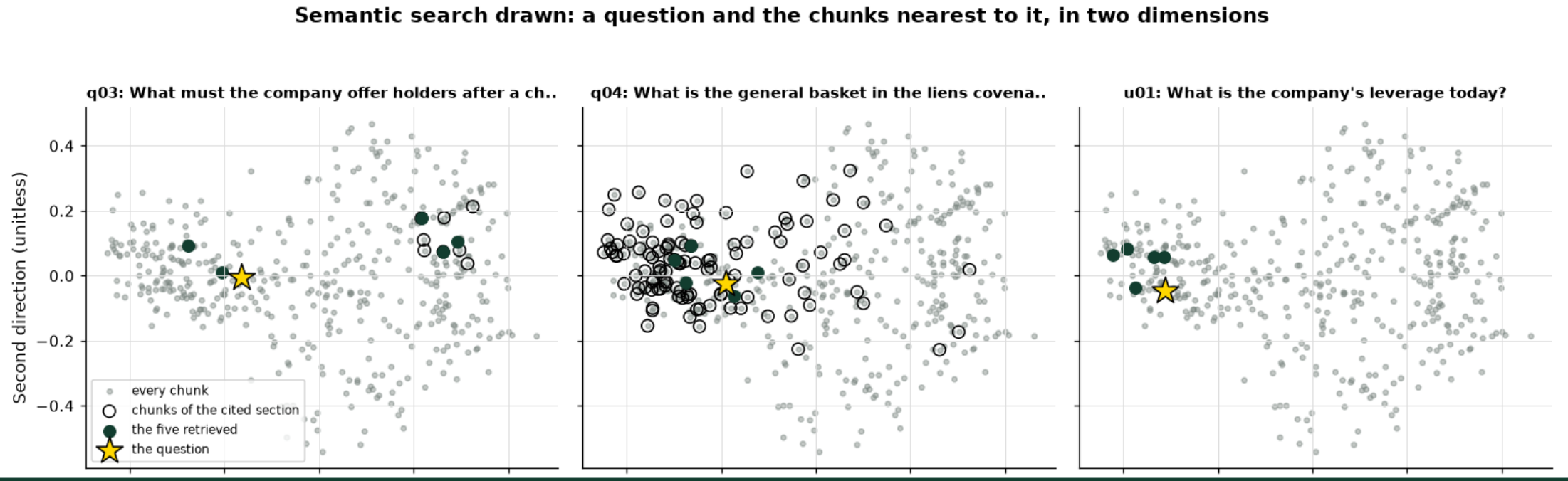
A vector database earns its place at millions of vectors, with documents added and removed all day, filters, and many users. None of that is here.

No framework either: the course checks the exact text sent, the text returned, and the check between.

How the settings of retrieval trade off

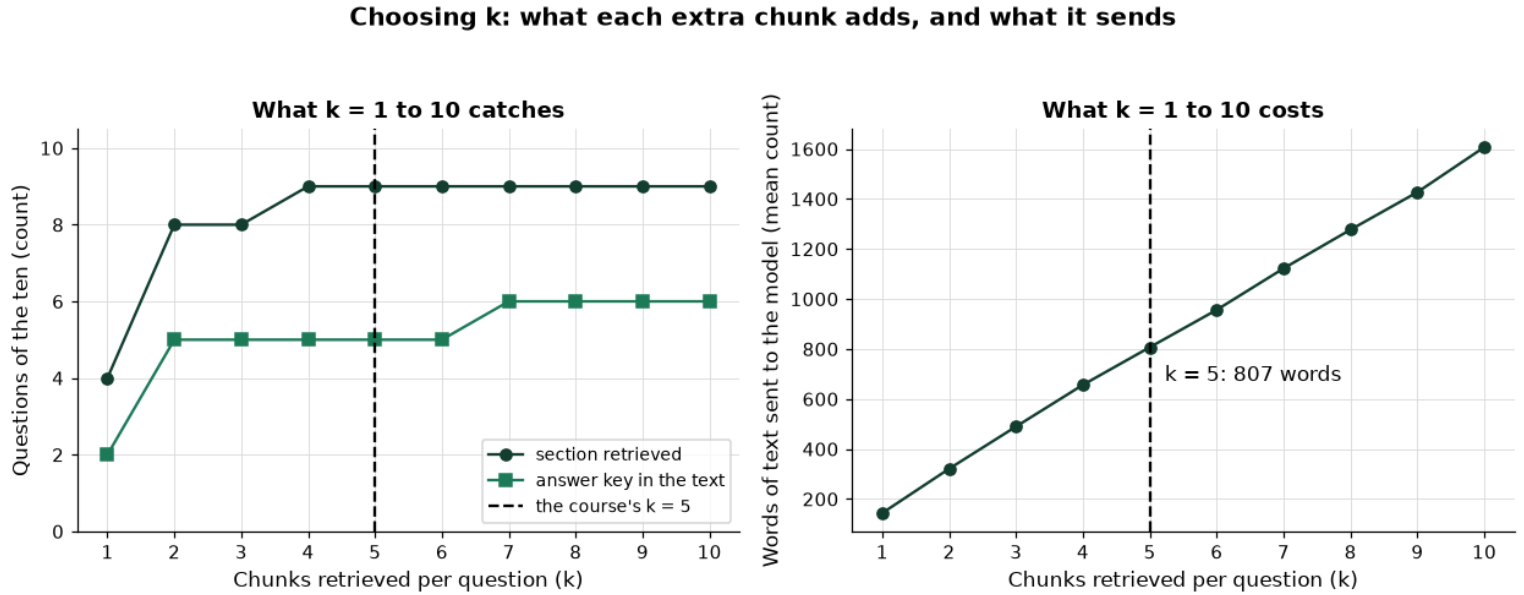
The problem	Each setting buys one thing and costs another
Chunk size	Short: sharp match, little context. Long: context, diluted match
Overlap	Keeps a sentence on a boundary whole
k	Small can miss the section; large sends more text
What comes out	A recall by k and a prompt size to read before choosing

Semantic search, drawn




```
q03: top five sections ['1.01', '4.10', '4.17', '4.10', '12.14']
q04: top five sections ['7.11', '4.14', '4.14', '1.01', '1.01']
u01: top five sections ['1.01', '1.01', '4.15', '4.15', '4.16']
```

Choosing k



recall by section at k = 1, 3, 5, 10: 4, 8, 9, 9 of 10; answer key in the text at the same k: 2, 5, 5, 6 of 10

807

mean words sent at k = 5

Past the knee, each chunk sends text and catches no new section.

The chunker's settings

Setting	Chunks	Mean words	What it does
90 words, overlap 15	871	85.0	short chunks, sharper matches, more of them
180 words, overlap 0	399	156.9	no shared words: a sentence on a boundary is cut
180 words, overlap 30	454	160.8	the course's setting
360 words, overlap 60	257	278.6	long chunks, fewer, more text in each answer

Only the count is shown: each setting needs its own run of embed and its own recall to judge.

Excel: the retrieval table

Excel side by side (tested in Excel)

`=ISNUMBER(MATCH(B2,C2:G2,0))`	10 of 10 rows equal to Python
`=COUNTIF(H2:H11,TRUE)`	9: recall at 5
`=ISNUMBER(MATCH(B2,C2:G2,0))  C:G General, 4.10 entered`	FALSE: 4.10 stored as 4.1
`=ISNUMBER(MATCH(B2,C2:G2,0))  C:G entered as '4.10`	TRUE: kept as text

Format the section columns as Text first, or enter each with an apostrophe.

Git: extractions.csv, 12 rows, 9 with both checks; then git tag week3 on the commit.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Units, conventions, and what the function must never do: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the judge looks for the answer's numbers in the whole indenture, so a number never retrieved passes.

Code that runs is not the same as code that is right.

What you can do now

- Send the chunks that matter, not the whole indenture
- retrieve with sections; rag_messages asks for citations
- Decide the questions and answer keys before any request
- judge_answer checks grounding: grounded is not right
- Count NOT IN TEXT apart; tag a commit in Git

Week 3, Day 5 The capstone: a covenant tool